

Continuous Random Variables and Probability Distributions

- **4.1 Continuous Random Variables and Probability Density Functions**
- **4.2 Cumulative Distribution Functions and Expected Values**
- **4.3 The Normal Distribution**
- **4.4 The Gamma Distribution and Its Relatives**
- **4.5 Other Continuous Distributions**
- **4.6 Probability Plots**

4.3 The Normal Distribution

■ Normal (Gaussian) Distribution

A continuous rv X is said to have a normal distribution with parameters μ and σ (or μ and σ^2), where $-\infty < \mu < +\infty$ and $0 < \sigma$, if the pdf of X is

$$f(x; \mu, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2 / (2\sigma^2)} \quad -\infty < x < \infty$$

Note:

1. The normal distribution is the most important one in all of probability and statistics. Many numerical populations have distributions that can be fit very closely by an appropriate normal curve.
2. Even when the underlying distribution is discrete, the normal curve often gives an excellent approximation.
3. Central Limit Theorem (see next Chapter)

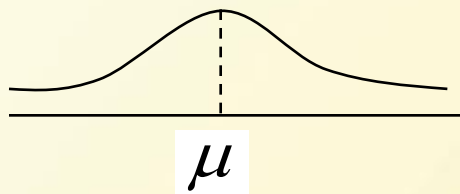
4.3 The Normal Distribution

■ Properties of $f(x;\mu,\sigma)$

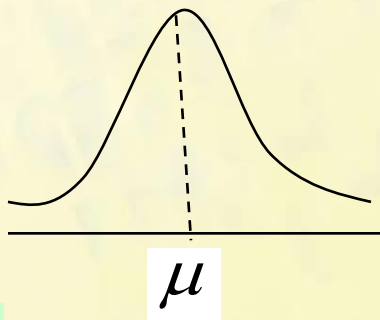
$$f(x;\mu,\sigma) \geq 0, \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}\sigma} e^{-(x-\mu)^2/(2\sigma^2)} dx = 1 \quad \text{Proof?}$$

$$E(X) = \mu \quad \& \quad V(X) = \sigma^2 \quad , \quad X \sim N(\mu, \sigma^2)$$

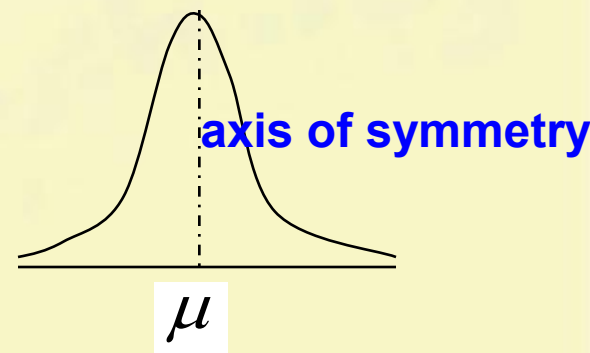
σ is large



σ is medium



σ is small



μ is location of axis of symmetry
 σ control the shape of the graph

Symmetry Shape

4.3 The Normal Distribution

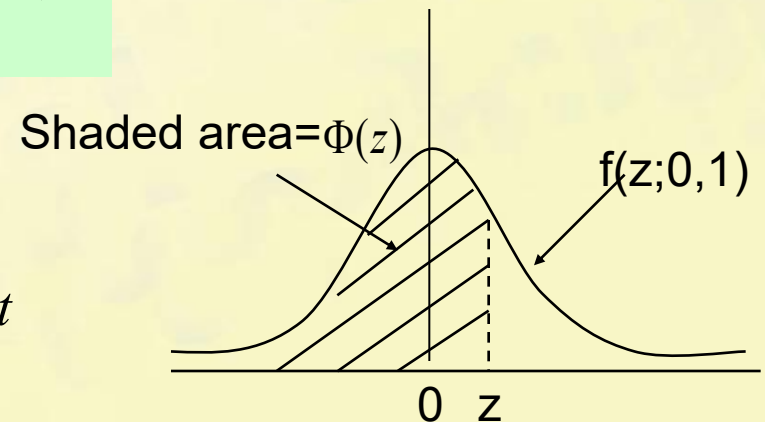
■ Standard Normal Distribution

The normal distribution with parameter values $\mu=0$ and $\sigma=1$ is called the **standard normal distribution**. A random variable that has a standard normal distribution is called a standard normal random variable and will be **denoted by Z**. **The pdf of Z is**

$$f(z;0,1) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}, \quad -\infty < z < \infty$$

The cdf of Z is

$$\Phi(z) = \int_{-\infty}^z f(t)dt = \int_{-\infty}^z \frac{1}{\sqrt{2\pi}} e^{-t^2/2} dt$$



Refer to Appendix Table A.3

4.3 The Normal Distribution

- Properties of $\Phi(z)$

$$\Phi(-z) = 1 - \Phi(z)$$

$$\Phi(0) = 0.5$$

$$P(|X| \leq z) = 2\Phi(z) - 1$$

$$P(|X| \geq z) = 2[1 - \Phi(z)]$$

4.3 The Normal Distribution

■ Example

Find $\Phi(1.65)$, $\Phi(-1.96)$ using **appendix Table A.3**

Solution:

Z	0.05	0.06
1.6	0.9505	
1.9		0.9750

$$\Phi(1.65) = 0.9505$$

$$\Phi(-1.96) = 1 - \Phi(1.96) = 1 - 0.9750 = 0.0250$$

Example

Given $X \sim N(0,1)$

Find $P(X \leq 1.65)$

$P(1.65 < X \leq 2.09)$

Solution:

$$\begin{aligned} P(X < 1.65) &= \Phi(1.65) = 0.9505 \\ P(1.65 < X < 2.09) &= \Phi(2.09) - \Phi(1.65) \\ &= 0.9817 - 0.9505 = 0.0312 \end{aligned}$$

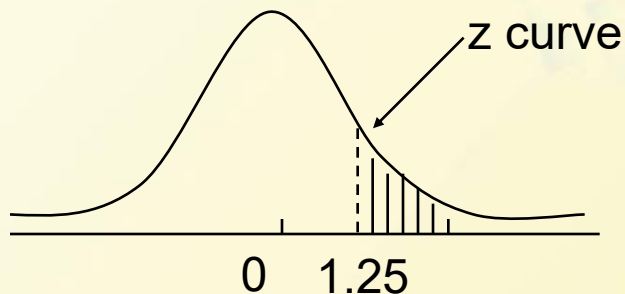
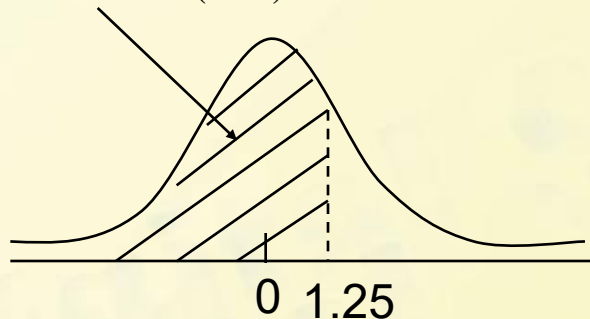
4.3 The Normal Distribution

■ Example 4.13

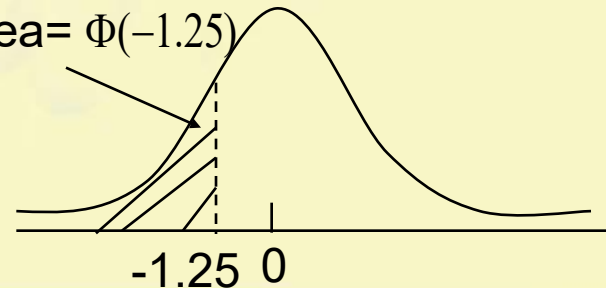
Compute the following standard normal probabilities:

- (a) $P(Z \leq 1.25)$ (b) $P(Z > 1.25)$ (c) $P(Z \leq -1.25)$

Shaded area = $\Phi(1.25)$ = 0.8944



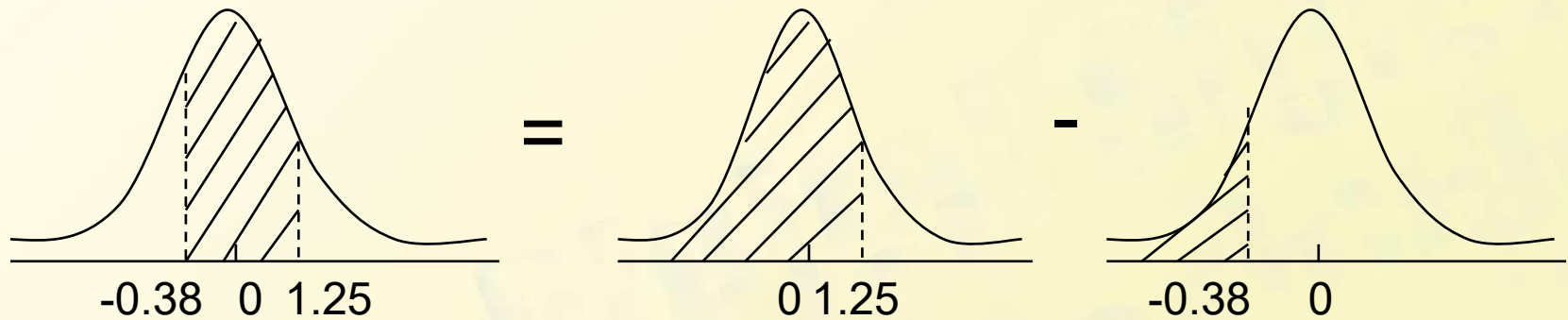
Shaded area = $\Phi(-1.25)$



4.3 The Normal Distribution

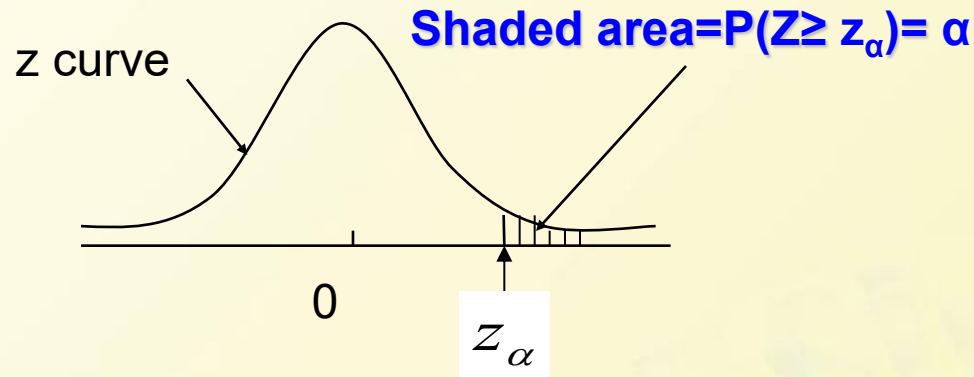
■ Example 4.13(Cont')

(d) $P(-0.38 \leq Z \leq 1.25)$



z_α notation

z_α will denote the values on the measurement axis for which α of the area under the z curve **lies to the right of z_α**



Note: z_α is the $100(1 - \alpha)$ th percentile of the standard normal distribution

Table 4.1 standard normal percentiles and critical values

Percentile	90	95	97.5	99	99.5	99.9	99.95
α (tail area)	0.1	0.05	0.025	0.01	0.005	0.001	0.0005
$z_\alpha = 100(1 - \alpha)$ th percentile	1.28	1.645	1.96	2.33	2.58	3.08	3.27

4.3 The Normal Distribution

■ Nonstandard Normal Distribution

If X has the **normal** distribution with **mean** μ and **standard deviation** σ , then

$$Z = \frac{X - \mu}{\sigma}$$

has a **standard normal distribution** (why?).

Relationship between **Nonstandard** Distribution and **Normal** Distribution

When $X \sim N(\mu, \sigma^2)$

$$F(x) = \frac{1}{\sqrt{2\pi}\sigma} \int_{-\infty}^x e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

If we let

$$\frac{x-\mu}{\sigma} = t, \quad x = \mu + \sigma t, \quad dx = \sigma dt$$

Then $F(x)$

$$\begin{aligned} &= \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\frac{x-\mu}{\sigma}} e^{-\frac{t^2}{2}} dt \\ &= \Phi\left(\frac{x-\mu}{\sigma}\right) \end{aligned}$$

Thus, when $X \sim N(\mu, \sigma^2)$

$$Y = \frac{X - \mu}{\sigma} \sim N(0, 1)$$

Proposition

If X has the normal distribution with mean μ and standard deviation σ , then

$$Z = \frac{X - \mu}{\sigma}$$

has a standard normal distribution. Thus

$$P(a \leq X \leq b) = P\left(\frac{a - \mu}{\sigma} \leq Z \leq \frac{b - \mu}{\sigma}\right) = \Phi\left(\frac{b - \mu}{\sigma}\right) - \Phi\left(\frac{a - \mu}{\sigma}\right)$$

$$P(X \leq a) = \Phi\left(\frac{a - \mu}{\sigma}\right)$$

$$P(X \geq b) = 1 - \Phi\left(\frac{b - \mu}{\sigma}\right)$$

Example

Given $X \sim N(3, 2^2)$, **find** $P(X > 2)$

Solution:

$$\text{If } X > 2, \text{ then } Y = \frac{X - \mu}{\sigma} = \frac{X - 3}{2} > -\frac{1}{2}$$

$$P(X > 2) = P(Y > -\frac{1}{2})$$

$$Y \sim N(0, 1)$$

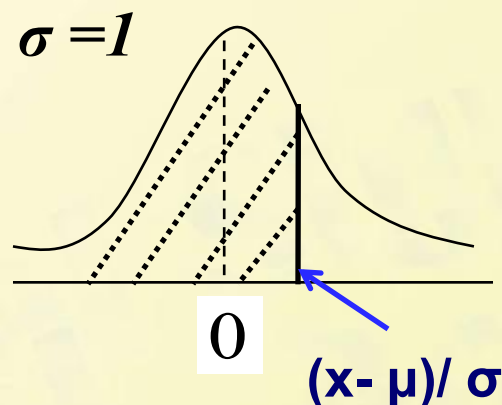
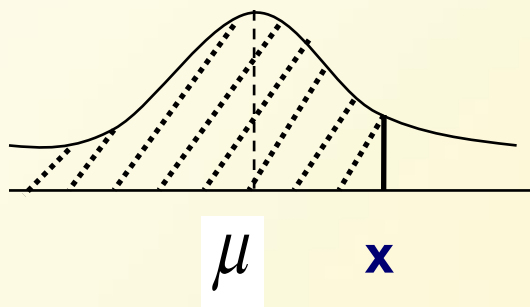
$$= 1 - P(Y \leq -\frac{1}{2}) = 1 - \Phi(-\frac{1}{2})$$

$$= \Phi(\frac{1}{2}) = 0.6915$$

4.3 The Normal Distribution

- Equality of nonstandard and standard normal curve area

$$P(Z \leq z) = P(X \leq \sigma z + \mu) = \int_{-\infty}^{\sigma z + \mu} f(x; \mu, \sigma) dx$$



Percentiles of an Arbitrary Normal Distribution

$$\begin{aligned} & (100p)\text{th percentile for normal } (\mu, \sigma) \\ &= \mu + [(100p)\text{th for standard normal}] \cdot \sigma \quad \text{Refer to Ex. 4.17} \end{aligned}$$

4.3 The Normal Distribution

■ Example 4.16

The time that it takes a driver to react to the brake lights on a decelerating vehicle is critical in helping to avoid rear-end collisions. Reaction time for an in-traffic response to a brake signal from standard brake lights can be modeled with a **normal distribution** having mean value 1.25 sec and standard deviation of .46 sec. What is the probability that reaction time is between 1.00 sec and 1.75 sec?

4.3 The Normal Distribution

Solution:

$$\begin{aligned}P(1.00 \leq X \leq 1.75) &= P\left(\frac{1.00 - 1.25}{0.46} \leq Z \leq \frac{1.75 - 1.25}{0.46}\right) \\&= \Phi(1.09) - \Phi(-0.54) \\&= 0.8621 - 2.946 = 0.5675\end{aligned}$$

4.3 The Normal Distribution

■ Example 4.17

The breakdown voltage of a randomly chosen diode of a particular type is known to be **normally distributed**.

What is the probability that a diode's breakdown voltage is **within 1 standard deviation of its mean value**?

4.3 The Normal Distribution

Solution:

$P(X \text{ is within 1 standard deviation of its mean})$

$$= P(\mu - \sigma \leq X \leq \mu + \sigma) = P\left(\frac{\mu - \sigma - \mu}{\sigma} \leq Z \leq \frac{\mu + \sigma - \mu}{\sigma}\right)$$

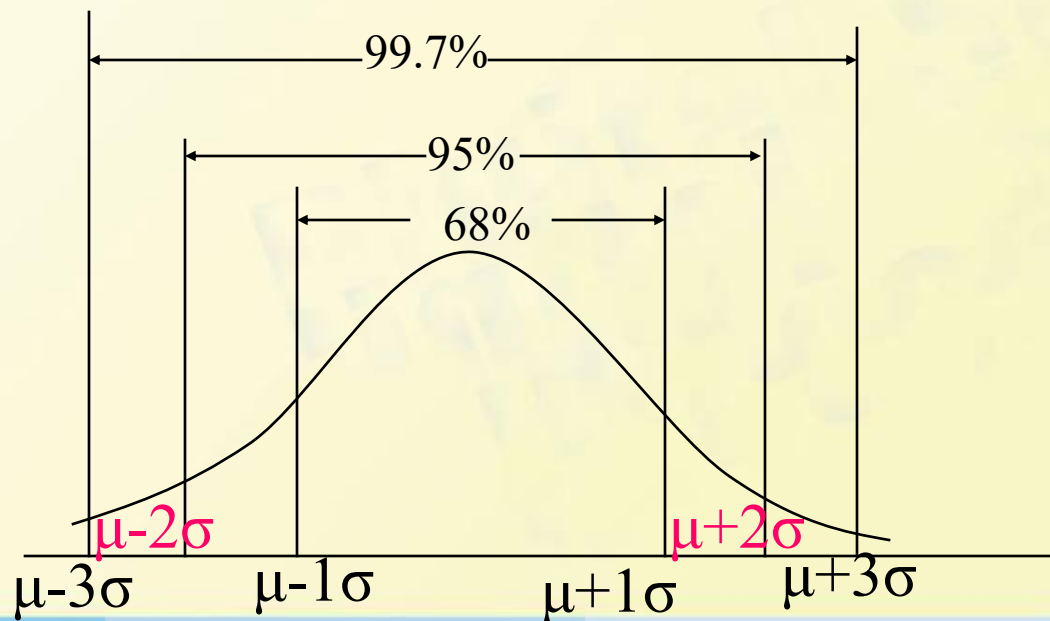
$$= P(-1.00 \leq Z \leq 1.00) = \Phi(1.00) - \Phi(-1.00) = 0.6826$$

Note: This question can be answered without knowing either μ or σ , as long as the distribution is known to be normal; in other words, the answer is the same for any normal distribution:

4.3 The Normal Distribution

If the population distribution of a variable is (approximately) normal, then

1. Roughly **68%** of the values are within **1 SD** of the mean.
2. Roughly **95%** of the values are within **2 SDs** of the mean
3. Roughly **99.7%** of the values are within **3 SDs** of the mean



4.3 The Normal Distribution

- The Normal Distribution and Discrete Populations

The **normal distribution** is often used as an approximation to the distribution of values in a **discrete population**.

Ex. 4.19:

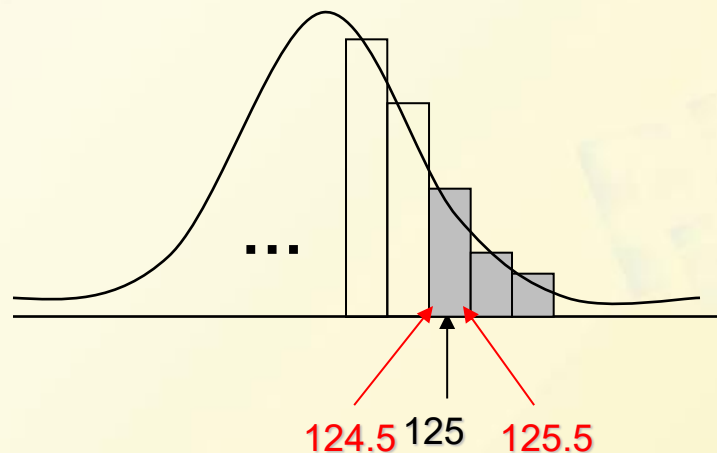
IQ in a particular population is known to be approximately normally distributed with $\mu = 100$ and $\sigma = 15$. What is the probability that a randomly selected individual has an IQ of **at least** 125?

The Normal Distribution and Discrete Populations

Solution:

Letting X = the IQ of a randomly chosen person, we wish $P(X \geq 125)$.

The temptation here is to standardize $X \geq 125$ immediately as in previous example. However, **the IQ population is actually discrete, since IQs are integer-valued, so the normal curve is an approximation to a discrete probability histogram,**



continuity correction

$$\begin{aligned} P(X \geq 125) &= P(Z \geq [(125 - 0.5) - 100] / 15) \\ &= P(Z \geq 1.63) = 0.0516 \end{aligned}$$

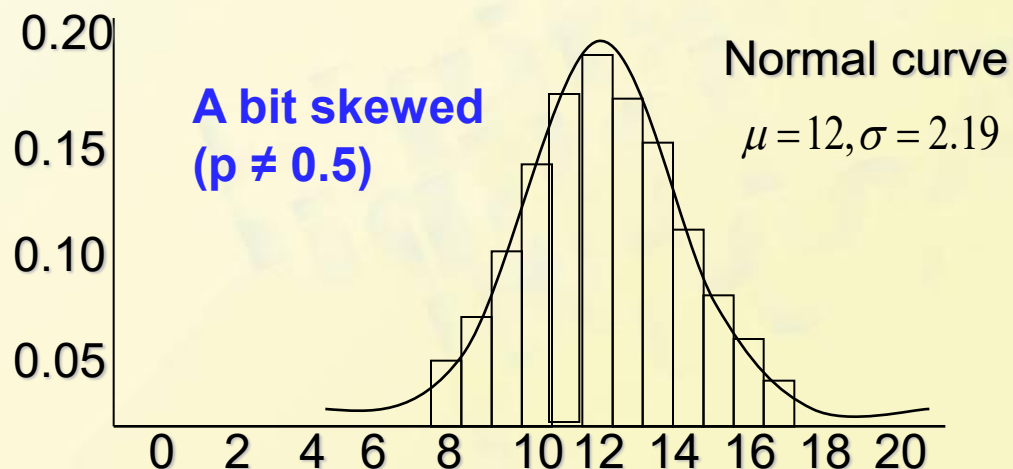
$$\begin{aligned} P(X = 125) &= P([(125 - 0.5) - 100] / 15 \leq Z \leq [(125 + 0.5) - 100] / 15) \\ &= P(1.63 \leq Z \leq 1.7) \neq 0 \end{aligned}$$

4.3 The Normal Distribution

- **The Normal Approximation to the Binomial Distribution**

Recall that the **mean value** and **standard deviation** of a **binomial random variable** X are $\mu_X = np$ and $\sigma_X = (npq)^{1/2}$.

Consider the binomial probability histogram with $n = 20$, $p = 0.6$. It can be approximated **by the normal curve** with $\mu = 12$ and $\sigma = 2.19$ as follows.



4.3 The Normal Distribution

■ Proposition

Let X be a binominal rv based on n trials with success probability p . Then if the binomial probability histogram is not too skewed, X has approximately a normal distribution with $\mu = np$ and $\sigma_X = (npq)^{1/2}$.

In particular, for $x =$ a possible value of X ,

$$\begin{aligned} p(X \leq x) &= B(x; n, p) \\ &\approx (\text{area under the normal curve to the left of } x + 0.5) \\ &= \Phi\left(\frac{x + 0.5 - np}{\sqrt{npq}}\right) \end{aligned}$$

Rule: In practice, the approximation is adequate provided that both $np \geq 10$ and $nq \geq 10$. (where $q = 1 - p$)

4.3 The Normal Distribution

■ Example 4.20

Suppose that 25% of all licensed drivers in a particular state do not have insurance. **Let X be the number of uninsured drivers in a random sample of size 50, so that $p=0.25$.**

- Find:**
- (A) $P(X \leq 10)$
 - (B) $P(5 \leq X \leq 15)$

Solution:

(A) Since $np=50(0.25)=12.5 \geq 10$ and $nq=37.5 \geq 10$, **the approximation can safely be applied**. Then $\mu = 12.5$ and $\sigma = 3.06$.

$$\begin{aligned} P(X \leq 10) &= B(10; 50, 0.25) \approx \Phi\left(\frac{10 + 0.5 - 12.5}{3.06}\right) \\ &= \Phi(-0.65) = 0.2578 \end{aligned}$$

(B) Similarly, the probability that between 5 and 15 (inclusive) of the selected drivers are uninsured is

$$\begin{aligned} P(5 \leq X \leq 15) &= B(15; 50, 0.25) - B(4; 50, 0.25) \\ &\approx \Phi\left(\frac{15.5 - 12.5}{3.06}\right) - \Phi\left(\frac{4.5 - 12.5}{3.06}\right) = 0.8320 \end{aligned}$$

4.4 The Gamma Distribution and Its Relatives

■ Gamma Function

For $\alpha > 0$, the gamma function $\Gamma(\alpha)$ is defined by

$$\Gamma(\alpha) = \int_0^{\infty} x^{\alpha-1} e^{-x} dx$$

The most important **properties** of the gamma function are the following:

1. For any $\alpha > 1$, $\Gamma(\alpha) = (\alpha-1) \Gamma(\alpha-1)$;
2. **For any positive integer n** , $\Gamma(n) = (n-1)!$
3. $\Gamma(1/2) = \sqrt{\pi}$

4.4 The Gamma Distribution and Its Relatives

■ The Family of Gamma Distributions

A continuous random variable X is said to have a gamma distribution if the pdf of X is

$$f(x; \alpha, \beta) = \begin{cases} \frac{1}{\beta^\alpha \Gamma(\alpha)} x^{\alpha-1} e^{-x/\beta} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

where the parameters α and β satisfy $\alpha > 0, \beta > 0$.

4.4 The Gamma Distribution and Its Relatives

- Mean and Variance

The mean and variance of a random variable X having the gamma distribution $f(x;\alpha,\beta)$ are

$$E(X) = \mu = \alpha\beta$$

$$V(X) = \sigma^2 = \alpha\beta^2$$

4.4 The Gamma Distribution and Its Relatives

■ Standard Gamma Distribution

The standard gamma distribution has $\beta = 1$.

$$f(x; \alpha) = \begin{cases} \frac{x^{\alpha-1} e^{-x}}{\Gamma(\alpha)} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

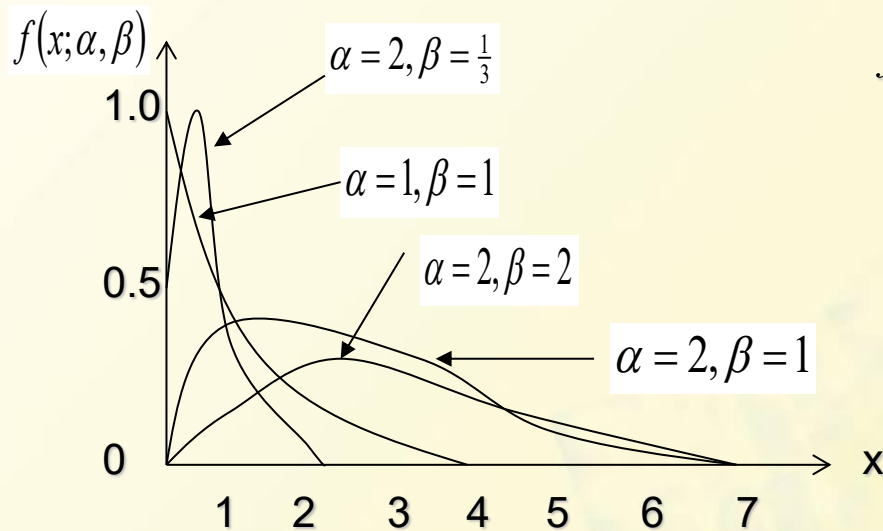
Satisfying the two Basic Properties of a pdf:

$$1: f(x; a) \geq 0$$

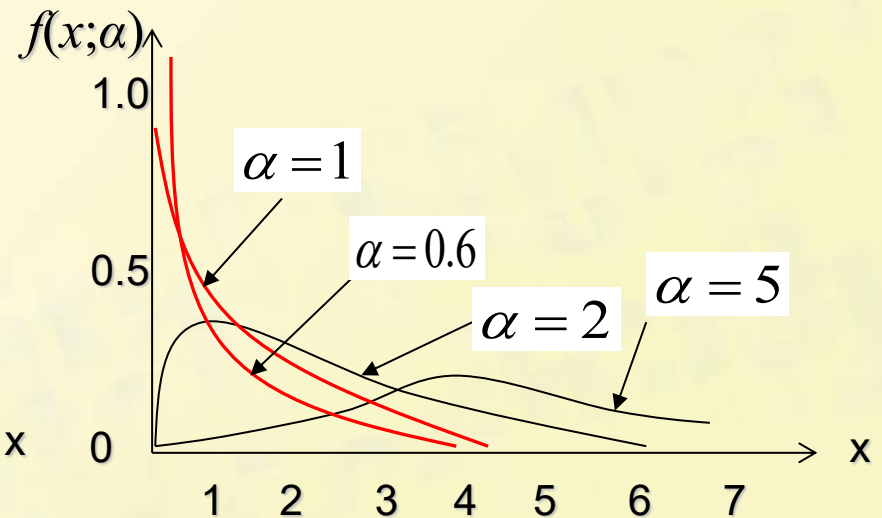
$$2: \int_0^{\infty} f(x; a) dx = \frac{\int_0^{\infty} x^{\alpha-1} e^{-x} dx}{\Gamma(\alpha)} = \frac{\Gamma(\alpha)}{\Gamma(\alpha)} = 1$$

4.4 The Gamma Distribution and Its Relatives

- Illustrations of the Gamma pdfs



(a) Gamma density curves



(b) Standard gamma density curves

4.4 The Gamma Distribution and Its Relatives

- The cdf of a standard gamma distribution

$$F(x; \alpha) = \int_0^x \frac{y^{\alpha-1} e^{-y}}{\Gamma(\alpha)} dy \quad x > 0$$

Incomplete gamma function (or without the denominator $\Gamma(\alpha)$ sometimes)

Refer to Appendix Table A.4

4.4 The Gamma Distribution and Its Relatives

■ Example 4.22

Suppose the reaction time X of a randomly selected individual to a certain stimulus has a **standard gamma distribution** with $\alpha=2$ sec. Then

$$P(3 \leq X \leq 5)=?$$

$$P(X > 4)=?$$

4.4 The Gamma Distribution and Its Relatives

Solution:

$$\begin{aligned}P(3 \leq X \leq 5) &= F(5;2) - F(3;2) \\ &= 0.960 - 0.801 = 0.159\end{aligned}$$

$$P(X > 4) = 1 - P(X \leq 4) = 1 - F(4;2) = 1 - 0.908 = 0.092$$

4.4 The Gamma Distribution and Its Relatives

■ Proposition

Let X have a gamma distribution with parameters α and β . Then for any $x > 0$, the **cdf** of X is given by

$$P(X \leq x) = F(x; \alpha, \beta) = F\left(\frac{x}{\beta}; \alpha\right)$$

where $F(\cdot; \alpha)$ is the **incomplete gamma function**.

4.4 The Gamma Distribution and Its Relatives

■ Example 4.24

Suppose the survival time X in weeks of a randomly selected male mouse exposed to 240 rads of gamma radiation has a **gamma distribution** with $\alpha=8$ and $\beta=15$, then

- (1) The probability that a mouse survives **between 60 and 120 weeks** is ?
- (2) The probability that a mouse survives **at least 30 weeks** is

4.4 The Gamma Distribution and Its Relatives

■ The Exponential Distribution

X is said to have an exponential distribution with parameter λ ($\lambda > 0$) if the **pdf** of X is

$$f(x; \lambda) = \begin{cases} \lambda e^{-\lambda x} & x \geq 0 \\ 0 & \text{otherwise} \end{cases}$$

Just a special case of the general gamma pdf

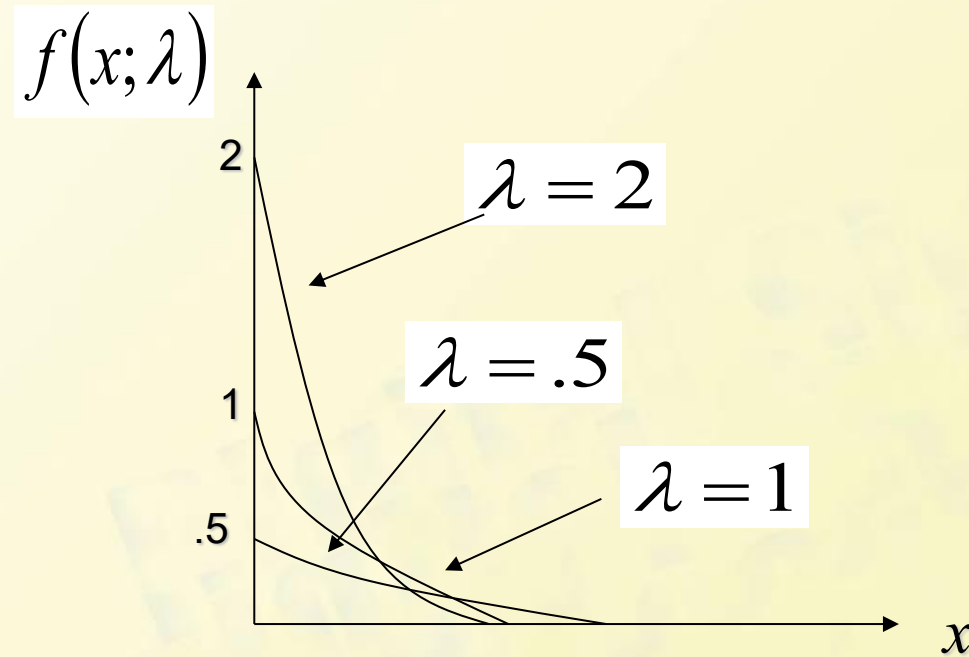
$$\alpha = 1 \text{ and } \beta = 1/\lambda$$

therefore, we have

$$E(X) = \alpha\beta = 1/\lambda; \quad V(X) = \alpha\beta^2 = 1/\lambda^2$$

4.4 The Gamma Distribution and Its Relatives

- Illustrations of the Exponential pdfs



4.4 The Gamma Distribution and Its Relatives

■ The cdf of Exponential Distribution

Unlike the general gamma pdf, the exponential pdf **can be easily integrated**.

$$F(x; \lambda) = \begin{cases} 0 & x < 0 \\ 1 - e^{-\lambda x} & x \geq 0 \end{cases}$$

4.4 The Gamma Distribution and Its Relatives

■ Example

Suppose the response time X at a certain on-line computer terminal (the elapsed time between the end of a user's inquiry and the beginning of the system's response to inquiry) has an **exponential distribution** with expected response time equal to **5 sec**. then $E(X) = 1/\lambda = 5$, so $\lambda = 0.2$. the probability that **the response time is at most 10 sec** is

$$P(X \leq 10) = F(10; 0.2) = 1 - e^{-(0.2)(10)} = 0.865$$

The probability that response time is **between 5 and 10 sec** is

$$P(5 \leq X \leq 10) = F(10; 0.2) - F(5; 0.2) = 0.233$$

4.4 The Gamma Distribution and Its Relatives

■ The Chi-Squared Distribution

Let ν be a **positive integer**. Then a random variable X is said to have a chi-squared distribution with parameter ν if the **pdf of X** is the **gamma density with $\alpha = \nu/2$ and $\beta = 2$** . The pdf of a chi-squared rv is thus

$$f(x, \nu) = \begin{cases} \frac{1}{2^{\nu/2} \Gamma(\nu/2)} x^{(\nu/2)-1} e^{-x/2} & x \geq 0 \\ 0 & x < 0 \end{cases}$$

The parameter ν is called **the number of degrees of freedom of X** . The symbol χ^2 is often used in place of “chi-squared.”